



9993 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRSPEC IFU				
	1	NIRSPEC G395H IFU	NIRSpec IFU Spectroscopy	(1) WISPIT-2
	5	NIRSPEC G395H IFU	NIRSpec IFU Spectroscopy	(1) WISPIT-2

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCAM BAR				
	2	WISPIT2-ROLL1	NIRCam Coronagraphic Imaging	(1) WISPIT-2
	3	WISPIT2-ROLL2	NIRCam Coronagraphic Imaging	(1) WISPIT-2
	4	WISPIT2-PSF-REF	NIRCam Coronagraphic Imaging	(2) WISPIT2-PSF-REF

ABSTRACT

We propose to use NIRCam and NIRSpec to characterize the atmosphere and circumplanetary disk of the 5Myr old directly imaged protoplanet, WISPIT 2b. Of the dozens of protoplanets hypothesized to lurk within the annular gaps of circumstellar disks, WISPIT 2b is only the third to be directly imaged, and the first that is amenable to direct spectroscopy with JWST. We propose to use NIRCam and NIRSpec to characterize WISPIT 2b's accreting circumplanetary disk and atmosphere. We will (i) directly test hydrodynamic simulations of planet formation, (ii) place constraints on planetary magnetic fields at unprecedentedly young ages, and (iii) anchor formation models by measuring the composition of a still-forming planet with respect to circumstellar snowlines.

OBSERVING DESCRIPTION

We seek to measure the 3-5 micron spectral energy distribution of the accreting protoplanet WISPIT 2b with NIRCam (to obtain photometry in a series of medium band filters), and NIRSpec (to obtain a continuum-subtracted spectrum in the G395H mode).

We based our NIRCam observing plan and exposure time request on GTO 1412 and GO 6905 that used the LWBAR and the narrow fiducial point override to achieve high throughput at close separations in long wavelength filters. We simulated the scene with pyNRC to determine optimal readout parameters and exposure times to achieve our stated science goals. We anticipate detecting the planet at 5-10 sigma in each filter at a separation of $\sim 0.3''$, depending on the amount of excess emission from the circumplanetary disk. We used the JWST ETC to determine target acquisition readout and exposure times.

We based our NIRSpec observing plan and exposure time request on GTO 1188 and GO 5342 that used the IFU in the G395H mode to detect H₂S in the atmospheres of the closely separated giant planets HR 8799 e and AF Lep b (equivalent to WISPIT 2 b). The observing plan avoids complete saturation of the PSF beyond $0.1''$ while integrating efficiently enough to detect the planet at high confidence. The exposure time request is based on the ETC tool made as part of GTO 1414 (Ruffio et al. 2024) for high contrast spectroscopy in this mode.

Proposal 9993 - Targets - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems, Exoplanets, Pre-main sequence stars, Protoplanetary disks]				
	(2)	WISPIT2-PSF-REF	RA: 19 22 33.4675 (290.6394479d) Dec: -07 19 19.17 (-7.32199d) Equinox: J2000	Proper Motion RA: -16.475 mas/yr Proper Motion Dec: -5.036000084146508 mas/yr Parallax: 0.0088428" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[G dwarfs]				

Proposal 9993 - Observation 1 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Observation	Proposal 9993, Observation 1: NIRSPEC G395H IFU											Mon Apr 27 22:00:11 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 1:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Exoplanet Systems, Exoplanets, Pre-main sequence stars, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0		50.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	15	3	false	true	NONE	16	48	11204.268	

Proposal 9993 - Observation 1 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Special Requirements	Aperture PA Range 13.97164917 to 28.97164917 Degrees (V3 235.0 to 250.0) Aperture PA Range 73.97164917 to 83.97164917 Degrees (V3 295.0 to 305.0) Aperture PA Range 193.97164917 to 208.97164917 Degrees (V3 55.0 to 70.0) Aperture PA Range 253.97164917 to 263.97164917 Degrees (V3 115.0 to 125.0)
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Proposal 9993 - Observation 5 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Observation	Proposal 9993, Observation 5: NIRSPEC G395H IFU											Mon Apr 27 22:00:11 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Data Excess over middle threshold											
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 5:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000				Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Exoplanet Systems, Exoplanets, Pre-main sequence stars, Protoplanetary disks]											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift (deg)		Column shift (deg)		Tile Order	
	1	2	10.0		50.0		0.0		0.0		DEFAULT	
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		MEDIUM		1		16				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2RAPID	12	3	false	true	NONE	16	48	9103.467	
	2	G140H/F100LP	NRSIRS2RAPID	12	3	false	true	NONE	16	48	9103.467	

Proposal 9993 - Observation 5 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Special Requirements

Aperture PA Range 13.97164917 to 28.97164917 Degrees (V3 235.0 to 250.0)
Aperture PA Range 73.97164917 to 83.97164917 Degrees (V3 295.0 to 305.0)
Aperture PA Range 193.97164917 to 208.97164917 Degrees (V3 55.0 to 70.0)
Aperture PA Range 253.97164917 to 263.97164917 Degrees (V3 115.0 to 125.0)

Proposal 9993 - Observation 2 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Observation	Proposal 9993, Observation 2: WISPIT2-ROLL1									Mon Apr 27 22:00:11 GMT 2026		
	Diagnostic Status: Warning											
Observing Template: NIRCcam Coronagraphic Imaging												
Diagnostics	(WISPIT2-ROLL1 (Obs 2)) Warning (Form): The selected fiducial point is not a standard option for the instrument.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(WISPIT2-ROLL1 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections			Miscellaneous				
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000		Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Star Description=[Exoplanet Systems, Exoplanets, Pre-main sequence stars, Protoplanetary disks]											
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	SAME	F335M	FAINT	RAPID	33	1	1	1.708	273966		
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern			
	A		MASKLWB		true		SUB400X256ALWB		NONE			
Confirmation	#	Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time		Conf. Total Dithers
	1	RAPID		4		1		1		42.947		1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F210M	F410M	DEEP8	20	7	1	7	2894.583			
	2	F212N	F430M	DEEP8	20	10	1	10	4135.119			
	3	F212N	F460M	DEEP8	20	13	1	13	5375.655			
	4	F212N	F480M	DEEP8	20	15	1	15	6202.679			

Proposal 9993 - Observation 2 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

PSF References	WISPIT2-PSF-REF (Obs 4) (PSF Reference; Filters [F210M/F410M, F212N/F430M, F212N/F460M, F212N/F480M]) WISPIT2-ROLL2 (Obs 3) (Filters [F210M/F410M, F212N/F430M, F212N/F460M, F212N/F480M]) Additional Justification: false
Special Requirements	No Parallel Attachments Fiducial Point Override NRCA5_MASKLWB_NARROW Group Observations 2, 3, 4, Non-interruptible Aperture PA Offset 2 from 3 by 7 to 14 Degrees (Same offsets in V3)

Proposal 9993 - Observation 3 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Observation	Proposal 9993, Observation 3: WISPIT2-ROLL2									Mon Apr 27 22:00:11 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCcam Coronagraphic Imaging									
Diagnostics	(WISPIT2-ROLL2 (Obs 3)) Warning (Form): The selected fiducial point is not a standard option for the instrument. (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (WISPIT2-ROLL2 (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.									
Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections			Miscellaneous			
	(1)	WISPIT-2	RA: 19 23 17.0332 (290.8209717d) Dec: -07 40 55.08 (-7.68197d) Equinox: J2000 Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Exoplanet Systems, Exoplanets, Pre-main sequence stars, Protoplanetary disks]	Proper Motion RA: 6.308 mas/yr Proper Motion Dec: -27.137999904880417 mas/yr Parallax: 0.0074649" Epoch of Position: 2000						
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1	SAME	F335M	FAINT	RAPID	33	1	1	1.708	273966
Template	Module	Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A	MASKLWB		true		SUB400X256ALWB		NONE		
Confirmation	#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers			
	1	RAPID	4	1	1	42.947	1			
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F210M	F410M	DEEP8	20	7	1	7	2894.583	
	2	F212N	F430M	DEEP8	20	10	1	10	4135.119	
	3	F212N	F460M	DEEP8	20	13	1	13	5375.655	
	4	F212N	F480M	DEEP8	20	15	1	15	6202.679	

Proposal 9993 - Observation 3 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

PSF References	WISPIT2-PSF-REF (Obs 4) (PSF Reference; Filters [F210M/F410M, F212N/F430M, F212N/F460M, F212N/F480M]) WISPIT2-ROLL1 (Obs 2) (Filters [F210M/F410M, F212N/F430M, F212N/F460M, F212N/F480M]) Additional Justification: false
Special Requirements	No Parallel Attachments Fiducial Point Override NRCA5_MASKLWB_NARROW Group Observations 2, 3, 4, Non-interruptible Aperture PA Offset 2 from 3 by 7 to 14 Degrees (Same offsets in V3)

Proposal 9993 - Observation 4 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

Observation	Proposal 9993, Observation 4: WISPIT2-PSF-REF										Mon Apr 27 22:00:11 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Coronagraphic Imaging										
Diagnostics	(WISPIT2-PSF-REF (Obs 4)) Warning (Form): The selected fiducial point is not a standard option for the instrument.										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	WISPIT2-PSF-REF	RA: 19 22 33.4675 (290.6394479d)			Proper Motion RA: -16.475 mas/yr					
			Dec: -07 19 19.17 (-7.32199d)			Proper Motion Dec: -5.036000084146508 mas/yr					
			Equinox: J2000			Parallax: 0.0088428"					
						Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[G dwarfs]										
Acquisition	#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	SAME	F335M	FAINT	RAPID	33	1	1	1.708	273966	
Template	Module		Occulting Mask		Obtain Astrometric Confirmation Images?		Subarray		Dither Pattern		
	A		MASKLWB		true		SUB400X256ALWB		NONE		
Confirmation	#	Conf. Readout Pattern		Conf. Groups/Int	Conf. Integrations/Exp		Conf. Total Integrations	Conf. Total Exposure Time		Conf. Total Dithers	
	1	RAPID		4	1		1	42.947		1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F210M	F410M	DEEP8	20	7	1	7	2894.583		
	2	F212N	F430M	DEEP8	20	10	1	10	4135.119		
	3	F212N	F460M	DEEP8	20	13	1	13	5375.655		
	4	F212N	F480M	DEEP8	20	15	1	15	6202.679		

Proposal 9993 - Observation 4 - Measuring the composition and circumplanetary environment of a gap-clearing protoplanet

PSF References	PSF Reference: true
Special Requirements	No Parallel Attachments Fiducial Point Override NRCA5_MASKLWB_NARROW Group Observations 2, 3, 4, Non-interruptible